

Mobasshir Bhuiya Shagor

Dhaka, Bangladesh · +880 1682 834837 · mobasshir.bhuiya@gmail.com · mobasshirbhuiya.com
[Google Scholar](https://scholar.google.com/citations?user=94m0buiy) · github.com/bhuiyanmobasshir94 · [linkedin.com/in/mobasshir-bhuiya-shagor](https://www.linkedin.com/in/mobasshir-bhuiya-shagor)

RESEARCH INTERESTS

Evaluation instruments for personalization systems under distribution shift: longitudinal benchmarks, protocols that separate identity leakage from genuine generalisation, and audits of explanation faithfulness in recommenders. The questions come from nearly eight years of production machine learning. Longer term, I intend to carry the same measurement problem to agentic systems, where a model acts on a person's behalf rather than only advising them.

EDUCATION

American International University Bangladesh

B.Sc. in Computer Science & Engineering CGPA 3.49 / 4.00

Dhaka, Bangladesh

May 2014 – Dec 2018

National University of Singapore

Specialist Training & Prototype Development Programme, Artificial Intelligence

Singapore

Jun – Sep 2020

PUBLICATIONS

Peer-reviewed

1. **M. B. Shagor**, M. Z. H. Alvi, and K. T. Hasan. "CID: Cow Images Dataset for Regression and Classification." *ICCA '22: Proceedings of the 2nd International Conference on Computing Advancements*, March 2022. doi:10.1145/3542954.3543018
2. **M. B. Shagor**, M. Al Rayhan, K. I. Huq, I. Nahar, and K. T. Hasan. "CAM: Categorized Affect Map. Implementation of a Smart Geographic Information System for Categorizing Places Based on People's Affective Responses." *ICCA 2020: Proceedings of the International Conference on Computing Advancements*, January 2020. doi:10.1145/3377049.3377074

Under review and in preparation

3. "BoviShift: A Six-Year Multi-Attribute Audit of Visual Cattle Phenotyping." *Under review*, 2026. A longitudinal benchmark of 2,657 unique animals (2021–2026) with a three-way protocol decomposition separating identity leakage from temporal distribution shift.
4. "Explanation Faithfulness, Hallucination, and Fairness in LLM-Based Recommenders." *In preparation*. A reproducible audit protocol that runs without GPU compute.

RESEARCH EXPERIENCE

BoviShift Research Collaboration

Supervised by Prof. Dr. Saef Ullah Miah (AIUB), with Nilambar Halder Tonmoy (Indiana University)

Dhaka, Bangladesh

Ongoing

- Co-developed **BoviShift**, a six-year longitudinal phenotyping benchmark. A random split conflates recognising a previously seen animal with generalising to a new one; the benchmark's identity-disjoint and temporally forward protocols separate the two, so an accuracy drop can be attributed rather than only reported. The same decomposition applies to any benchmark built from repeat observations of the same entities, which describes most personalization data.
- The 2024 cohort is held out of the primary temporal arm because its missingness is non-random; including it would let a collection artifact pass as distribution shift.
- Designing a zero-GPU audit of LLM-based recommenders that tests whether stated explanations describe the actual basis of a recommendation, and whether failures fall unevenly across groups.

National University of Singapore

Prototype Development Programme: KhaoDao Recommendation Engine

Singapore

Jun – Sep 2020

- Led the design of a context-aware recommender using weather, day of week and time of day. Moved from classical collaborative filtering (Surprise, Implicit) to DLRM for better cold-start and sparsity behaviour, then optimised it for low-latency online inference.

AIUB Data Science Lab

Research Assistant, lab of Prof. Khandaker Tabin Hasan

Dhaka, Bangladesh

Sep 2017 – Dec 2018

- Developed the **Categorized Affect Map**, a category-affect-space model that classifies places by problem category from crowdsourced affective responses using multi-label classification. Led the thesis group through hackathon placement and publication, and led experimentation on the **Bengali WordNet** linguistic resource.

- **CID: Cow Images Dataset.** 17,899 images annotated across ten attributes, including breed, age, height and weight, released with regression and classification baselines. github.com/bhuiyanmobasshir94/CID
- **BoviShift.** 2,657 unique animals photographed 2021–2026 in a single marketplace, with the three-protocol evaluation harness. To be released with the paper.
- **Margin.** Open-source adaptive learning system: two-stage mastery, spaced review, and error patterns from a learner's last three graded essays pulling matching lessons forward. An LLM does the reasoning; deterministic code bounds every output, clamping scores to the legal lattice, recomputing aggregates from their components, and treating provider output as untrusted input. github.com/bhuiyanmobasshir94/margin

PROFESSIONAL EXPERIENCE

Graaho Technologies

Dhaka, Bangladesh

Staff Machine Learning Engineer

2023 – present

- **AlgoRec:** multi-tenant recommendation platform. Every candidate model was scored by AUC against a popularity baseline before release. Implicit feedback modelled with ALS and confidence weighting; precomputed embeddings served through a 64-dimensional approximate nearest-neighbour index; every run versioned in MLflow with exactly one marked servable.
- **Koronik:** agent platform deployed for Islami Bank Bangladesh PLC across four channels, serving 5,000+ daily users. Responses below a confidence threshold are refused and escalated to human agents; tenant isolation is enforced at the database partition key, not in the prompt.
- **Smart Guess (CowNet):** PyTorch multi-input-output network predicting age, height, length, weight and breed from livestock images; used for initial procurement screening at Bengal Meat.
- **Easymator:** construction takeoff from architectural drawings, with scale calibration validated before any pixel-to-dimension conversion. Automatic wall detection was deliberately left out in favour of human tracing, because its errors would be invisible to the user.

Senior Machine Learning Engineer, seconded to Renforce, Sydney

Oct 2021 – Jan 2023

- Built a text-cluster classifier that structures regulatory PDF content into obligations verified against an ontology before ingestion, a named-entity metadata extractor, and a rule recogniser. Designed the ML lifecycle platform and a Kafka-driven workflow manager generating Airflow DAGs.

Machine Learning Engineer

Mar 2020 – Sep 2021

- Took the DLRM-based KhaoDao recommender to production with low-latency online prediction; built Bengali/English OCR services for identity verification; prototyped credit-risk models.

Data Scientist

Mar 2019 – Feb 2020

- Built the original Smart Guess pipeline: Mask R-CNN instance segmentation to extract girth keypoints, and a Keras multi-input-output network predicting height, length, weight and breed. *BASIS National ICT Award 2019.*

HONOURS AND AWARDS

- **BASIS National ICT Award**, second runner-up, AI in Agriculture, for Smart Guess. 2019
- **Kaggle 3× Expert**, top 5% of data scientists across notebooks, datasets and discussion. 2021
- **Specialist Training in AI**, Singapore e-Government Leadership Centre and Bangladesh Ministry of ICT. 2020
- **Top 10**, Sustainable Development Goal Hackathon, Banglalink Digital and a2i. 2018

TECHNICAL SKILLS

Research: PyTorch, TensorFlow, scikit-learn, OpenCV; instance segmentation, matrix factorisation and implicit feedback, sequence labelling, multi-label classification, retrieval-augmented generation. **Systems:** Python, C++, Java, SQL; FastAPI, Django; MLflow, Airflow, Celery, Ray; PostgreSQL, Docker. **Cloud:** AWS (SageMaker, Bedrock, Lambda, Glue, EMR), Spark, Kafka, Linux. **Certification:** AWS Certified Solutions Architect, Associate, 2024.

SELECTED TALKS

- “CID: Cow Images Dataset for Regression and Classification.” ICCA 2022. 2022
- “KhaoDao Recommendation Engine.” Prototype presentation, National University of Singapore. 2020